Operator Informed Hands on

Start of Block: Screenning Questions

Q4 Are you 18 years old or older?

- ○ Yes (1)
- ○ No (2)

Q1 How many years of experience do you have in satellite mission operations?

- ○ None (1)
- ○ 1 year or less (2)
- ○ 1 to 3 years (3)
- ○ 3 to 5 years (4)
- ○ 5 to 10 years (5)

Q3 How many years of experience do you have in cybersecurity?

- ○ None (1)
- ○ 1 year or less (2)
- ○ 1 to 3 years (3)
- ○ 3 to 5 years (4)
- ○ 5 to 10 years (5)

End of Block: Screenning Questions

End of Block: Consent

Start of Block: Demographic Questionnaire

Q9 What is your gender?

- ○ Female (1)
- ○ Male (2)
- ○ Other (please specify) (3) _____
- ○ Prefer not to answer (4)

Q10 How old are you?

- ○ 18-24 (1)
- ○ 25-34 (2)
- ○ 35-44 (3)
- ○ 45-54 (4)
- ○ 55-64 (5)
- ○ 65 or more (6)
- ○ Prefer not to answer (7)

Q11 What is the highest level of education you have completed?

- ○ High School (1)
- ○ Associate degree (2)
- ○ Bachelors degree (3)
- ○ Masters degree (4)
- ○ Ph.D. (5)
- ○ Prefer not to answer (6)

Q12 What is your first language?

- ○ English (1)
- ○ German (2)
- ○ Italian (3)
- ○ French (4)
- ○ Spanish (5)
- ○ Portuguese (6)
- ○ Chinese (Mandarin) (7)
- ○ Arabic (8)
- ○ Other (please specify) (9) _____
- ○ Prefer not to answer (10)

Q13 Where do you reside?

- ○ North America (1)
- ○ South America (2)
- ○ Europe (3)
- ○ Asia (4)
- ○ Africa (5)
- ○ Australia (6)
- ○ Caribbean Islands (7)
- ○ Pacific Islands (8)
- ○ Other (please specify) (9) _____
- ○ Prefer not to answer (10)

End of Block: Demographic Questionnaire

Start of Block: Professional Experience Questionnaire

Q14 What area are you currently employed in?

- ○ Industry (1)
- ○ Government (2)
- ○ Academia (3)
- ○ Other (please specify) (4) _____
- ○ Prefer not to say (5)

Q18 For each of the following cybersecurity skills, self-report your level of proficiency.

	Poor (1)	Average (2)	Good (3)	Very Good (4)	Excellent (5)
Network reconnaissance (1)	○	○	○	○	○
Network security (2)	○	○	○	○	○
Binary analysis/reverse engineering (3)	○	○	○	○	○
Penetration testing (4)	○	○	○	○	○
Embedded systems security (5)	○	○	○	○	○

Q20 For each of the following satellite-related skills, self-report your level of proficiency.

	Poor (1)	Average (2)	Good (3)	Very Good (4)	Excellent (5)
Satellite operations (1)	○	○	○	○	○
Satellite hardware engineering (2)	○	○	○	○	○
Satellite software engineering (3)	○	○	○	○	○
Orbital Mechanics/Physics (4)	○	○	○	○	○
Radio Operation (5)	○	○	○	○	○

End of Block: Professional Experience Questionnaire

Start of Block: Satellite Mission Experience

Q49 What is primary purpose or mission of the satellite(s) that you operate?

- ☐ Weather (1)
- ☐ Scientific Research (2)
- ☐ Earth Observation (3)
- ☐ Communications and Navigation (4)
- ☐ Remote Sensing (5)
- ☐ TV or Radio Broadcasting (9)
- ☐ Early Warning Systems (8)
- ☐ I do not know (6)
- ☐ Other (please specify) (10) _____
- ☐ Prefer not to say (7)

Q46 What is the type of orbit of the satellite(s) that you operate?

- ☐ Low Earth Orbit (LEO) (1)
- ☐ Medium Earth Orbit (MEO) (2)
- ☐ Geostationary Earth Orbit (GEO) (3)
- ☐ I do not know (4)
- ☐ Prefer not to say (5)

Q47 What is the size of the satellite(s) that you operate?

- ☐ Nanosatellite (1-10 kg) (1)
- ☐ Microsatellite (10-100 kg) (2)
- ☐ Minisatellite (100-500 kg) (3)
- ☐ Medium-sized satellites (500-1,000 kg) (4)
- ☐ Large satellites (1,000 kg or more) (5)
- ☐ I do not know (6)
- ☐ Prefer not to say (7)

Q263 How many different satellite missions have you operated?

	1 2 3 4 5 6 6 7 8 9 10
Satellite missions ()	

Q262 Have you ever operated a CubeSat satellite mission?

- ○ Yes (1)
- ○ No (2)
- ○ I do not know (3)
- ○ Prefer not to say (4)

Q50 What is the level of automation of the satellite mission(s) that you operate?

- ☐ Fully automated (1)
- ☐ Mostly automated (2)
- ☐ Some operations are automated (3)
- ☐ A few operations are automated (4)
- ☐ No automation at all (5)
- ☐ I do not know (6)
- ☐ Prefer not to say (7)

Q316 When operating a satellite mission how do you interact with it?

- ○ On-site operation (1)
- ○ Remote operation (2)
- ○ Both (3)
- ○ Prefer not to say (4)

Q317 What remote operation tool or tools do you use to remotely operate the satellite mission?

- ☐ SSH (1)
- ☐ Telnet (2)
- ☐ VNC (RealVNC, UltraVNC, etc.) (3)
- ☐ Remote Desktop Protocol (RDP) (4)
- ☐ Remote Desktop Application (TeamViewer, NoMachine, etc.) (5)
- ☐ Web Browser (6)
- ☐ Custom application (7)
- ☐ Other (please specify) (8) _____
- ☐ I do not know (9)
- ☐ Prefer not to say (10)

Q259 What type of software does the ground segment of the satellite mission(s) that you operate use? (Select all that apply).

- ☐ Software exclusively developed for this mission(s) (1)
- ☐ Existing software developed by my institution used for multiple missions (2)
- ☐ Existing software developed by an outside institution (not commercial) (3)
- ☐ Existing commercial software (4)
- ☐ Existing open-source software (5)
- ☐ I do not know (6)
- ☐ Prefer not to say (7)

Q260 What type of software does the space segment of the satellite mission(s) that you operate use? (Select all that apply).

- ☐ Software exclusively developed for this mission(s) (1)
- ☐ Existing software developed by my institution used for multiple missions (2)
- ☐ Existing software developed by an outside institution (not commercial) (3)
- ☐ Existing commercial software (4)
- ☐ Existing open-source software (5)
- ☐ I do not know (6)
- ☐ Prefer not to say (7)

End of Block: Satellite Mission Experience

Start of Block: Explain Mission

Q442

University CubeSat Mission Overview

In this study you will interact with a simulation system that is **modeled after a real CubeSat satellite mission designed and operated by a university**. The mission includes a 3U CubeSat performing Earth observation and system telemetry operations using common ground-based tools. In this simulation, you will perform core operational tasks using **two main interfaces**:

Gpredict: This software is used for pass prediction and antenna tracking. It shows when the CubeSat will be in communication range and helps position the ground station antenna accordingly. You'll use it to simulate real-time tracking of the satellite during its orbit.

Custom Mission Control Software (MCS) Terminal: The university developed their own custom MCS as a command line application. You will use this command line terminal to issue commands to the MCS, to send telecommands to the satellite (e.g., schedule operations, request data) and to receive telemetry output such as temperature readings. This custom MCS is used by many real-world in-orbit CubeSat projects.

Page Break

Q444

Command Overview

As is the case with most satellite missions, this university satellite mission MCS and flight software were designed to implement multiple telecommands for different purposes. In the context of this study there are two types of commands: **ground segment and space segment commands**. Both types of commands have a basic syntax that includes the command name and the parameters necessary to execute the command (if any).

Ground segment commands: These commands execute different operations on the ground segment MCS only and do not communicate with the space segment directly.

An example of a ground segment command is "*fp_show"*. The *fp_show* command is short for "show flight plan" and it displays a queue of telecommands that are to be sent to the spacecraft at specific times, normally when a pass occurs. This command does not need any additional parameters to work. Below is an example of the output once the fp_show command is executed.

Space segment commands: These commands do communicate with the space segment or spacecraft directly via the radio link and are used to download the latest telemetry or change some setting on the spacecraft. Additionally, these commands can only be executed successfully when the satellite is passing over the ground station. When a space segment command reaches the satellite and is executed successfully you will see the message "**Command results: 1**." On the other hand, if the command fails it will show this message: "**Error sending command" and "Command result: 0**."

An example of a space segment command is *"1: com_ping 10".* This command sends ping messages to the satellite and succeds if there is a response. **The syntax "1:" means that this command is sent to the CubeSat** with ID 1, which is the only CubeSat in this particular mission.

Page Break

Q443

University CubeSat Satellite Tasks Instructions

In the following tasks you will be given access to our satellite mission simulation system designed to simulate a **3U CubeSat satellite mission developed by a university**.

The study will be conducted via **two windows**. One window is the **Qualtrics window**, where you are on right now where you will read the instructions for each task and answer a series of questions after you have completed the task. The second window is the **videoconferencing window** for interacting with the simulation. You will be given access to the mission simulation via the videoconferencing screen sharing feature. During the session, the researcher will share their screen with you and grant you remote control of the system. At the same time, you will keep this Qualtrics survey open in a separate window. Each task will be described here in the survey, and you will also be given instructions on when you should switch to the videoconferencing window. After completing each one, you will answer a set of questions about the task you just performed. You may switch between the simulation and this survey at your own pace using Alt+Tab (Windows) or Command+Tab (Mac). There is no time limit for completing tasks, but we recommend completing each one before moving on to the next. You are free to move between both windows at any time; however, we do ask you to focus on these two windows only during the survey session. **If you have any question or concern at any point to during the study do not hesitate to let the researcher know if you need clarification or assistance at any time, they are here to assist you.** When you are ready, click “Next” to begin the first task.

End of Block: Explain Mission

Start of Block: Note about time and date

Q500

Note on the University CubeSat Mission Date and Time

To ensure a consistent and fair experience for all participants, the satellite mission simulation you will interact with **has been set to a specific date and time that does not correspond to the current day**. This is intentional and it is necessary to align the satellite’s orbit and visibility window with each participant's availability and to allow them to perform the hands-on tasks during a pass. All participants will work with the **same conditions**, even though they may complete the study on different dates. **Please proceed as if the time and date shown are correct in the CubeSat mission context.**

Page Break

Q503

Screen Sharing Note

The researcher will now share their screen and give you access to the satellite simulation system.

End of Block: Note about time and date

Start of Block: Familiarize with Interface

Q445

Familiarization with the CubeSat Simulation System

Before beginning the main tasks, you will now have an opportunity to explore the CubeSat simulation system and **familiarize yourself with the interfaces** described previously. During this familiarization period, you may: Navigate the system using the terminal and type the "help" command and hit enter. Gpredict, you can try right clicking the satellite in world map. Ask the researcher any questions about system features

There are no questions for this step and its purpose is to help you get comfortable with the tools and layout before you begin the actual study tasks. When you feel ready to proceed, switch to the videoconferencing window.

After you have familiarized yourself with the system and are ready, click on the "Next" arrow button.

End of Block: Familiarize with Interface

Start of Block: Task 1: Satellite Pass Prediction

Q72

Mission Operation Task 1: Pass Prediction

In this task you will use **Gpredict** to identify when the **next pass** of the CubeSat will occur over the designated ground station. To accomplish this, follow these steps: Switch to the videoconferencing window (remeber that you can switch back and forth between this and the videoconferencing window at any time) In the Gpredict interface **right click on the CubeSat** shown in the world map. A new option list will appear, **click future passes**. In the MCS terminal, send the command "**1: com_ping 10**" to the CubeSat and watch what happens when the satellite is **not passing** over the ground station.

When you are ready click "Next" arrow to go this task's questions.

Page Break

Q271 Have you performed the *"**pass prediction**"* operation before? *(Task 1: Question 1 of 5)*

- ○ Yes, several times. (1)
- ○ Yes, a few times. (2)
- ○ No, never (3)

Page Break

Q272 I understood what was the satellite operation being performed. *(Task 1: Question 2 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q273 Please explain what elements were not clear.

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Q274 Please explain what elements allowed you to understand the operation.

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Page Break

Q281 The output text and data shown when performing the **"pass prediction"** operation resembles the output text and data of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 1: Question 3 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q282 Please explain what elements of the output text and data shown do not resemble the operation of a real CubeSat satellite mission.

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Q283 Please explain what elements of the output text and data shown resemble the operation of a real CubeSat satellite mission.

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Page Break

Q284 The Graphical User Interface (GUI) used in the **"pass prediction"** operation resemble the GUI used in a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 1: Question 4 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q285 Please explain what elements of the GUI do not resemble the operation of a real CubeSat satellite mission.

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Q286 Please explain what elements of the GUI resemble the operation of a real CubeSat satellite mission.

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Q278 Overall, the **"pass prediction"** operation resembles the operation of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 1: Question 5 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q446 Please explain what overall elements do not resemble the operation of a real CubeSat satellite mission.

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Q447 Please explain what overall elements most resemble the operation of a real CubeSat satellite mission.

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End of Block: Task 1: Satellite Pass Prediction

Start of Block: Task 2: Telecommand Scheduling

Q448

Mission Operation Task 2: Telecommand Scheduling

In this task you are going to schedule two telecommands to be sent to the CubeSat at specific times. One will be sent **before** the CubeSat pass and the second one **at the beginning** of the pass. To accomplish this, follow these steps: Switch to the videoconferencing window **Before pass**: Schedule a ping command to be execute three minutes before the pass In the MCS terminal write the following command: "***fp_set_cmd 07 07 2025 14 40 00 1 0 com_ping 1"*** **At beggining of pass:** In the MCS terminal, write the following command ""***fp_set_cmd 07 07 2025 14 44 00 1 0 com_ping 1***" To check that the commands worked and are queued now execute the command "***fp_show***"

When you are ready click "Next" arrow to go this task's questions.

Page Break

Q449 Have you performed the *"***telecommand scheduling***"* operation before? *(Task 2: Question 1 of 5)*

- ○ Yes, several times. (1)
- ○ Yes, a few times. (2)
- ○ No, never (3)

Page Break

Q450 I understood what was the satellite operation being performed. *(Task 2: Question 2 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q451 Please explain what elements were not clear.

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Q452 Please explain what elements allowed you to understand the operation.

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Page Break

Q453 The output text and data shown when performing the **"telecommand scheduling"** operation resembles the output text and data of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 2: Question 3 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q454 Please explain what elements of the output text and data shown do not resemble the operation of a real CubeSat satellite mission.

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Q455 Please explain what elements of the output text and data shown resemble the operation of a real CubeSat satellite mission.

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Q456 The telecommands used in the **"telecommand scheduling"** operation resemble the telecommands used in a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 2: Question 4 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q457 Please explain what elements of the telecommands do not resemble the operation of a real CubeSat satellite mission.

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Q458 Please explain what elements of the telecommands resemble the operation of a real CubeSat satellite mission.

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Q459 Overall, the **"telecommand scheduling"** operation resembles the operation of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 2: Question 5 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q460 Please explain what overall elements do not resemble the operation of a real CubeSat satellite mission.

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Q461 Please explain what overall elements most resemble the operation of a real CubeSat satellite mission.

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Page Break

End of Block: Task 2: Telecommand Scheduling

Start of Block: Task 3: Ping

Q486

Mission Operation Task 3: Ping Communication Test

In this task you are going to observe the results of the two telecommands that you scheduled in the previous task. The telecommands are going to send ping messages to the CubeSat before it is passing over the ground station and at the beggining of the pass. To comple this task do the following: Switch to the videoconferencing window. **Wait for the CubeSat** to get closer to the ground station as shown in Gpredit and look at the MCS terminal for when the scheduled telecommands are executed. **In the MCS terminal, observe the results** of the telecommands that you scheduled. Are the results between the first and second ping different? When you are ready click "Next" arrow to go this task's questions.

Page Break

Q487 Have you performed the *"**ping test**"* operation before? *(Task 2: Question 1 of 5)*

- ○ Yes, several times. (1)
- ○ Yes, a few times. (2)
- ○ No, never (3)

Page Break

Q488 I understood what was the satellite operation being performed. *(Task 2: Question 2 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q489 Please explain what elements were not clear.

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Q490 Please explain what elements allowed you to understand the operation.

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Page Break

Q491 The output text and data shown when performing the **"ping test"** operation resembles the output text and data of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 2: Question 3 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q492 Please explain what elements of the output text and data shown do not resemble the operation of a real CubeSat satellite mission.

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Q493 Please explain what elements of the output text and data shown resemble the operation of a real CubeSat satellite mission.

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Q494 The telecommands used in the **"ping test"** operation resemble the telecommands used in a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 2: Question 4 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q495 Please explain what elements of the telecommands do not resemble the operation of a real CubeSat satellite mission.

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Q496 Please explain what elements of the telecommands resemble the operation of a real CubeSat satellite mission.

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Page Break

Q497 Overall, the **"ping test"** operation resembles the operation of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 2: Question 5 of 5)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q498 Please explain what overall elements do not resemble the operation of a real CubeSat satellite mission.

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Q499 Please explain what overall elements most resemble the operation of a real CubeSat satellite mission.

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Page Break

End of Block: Task 3: Ping

Start of Block: Task 3: Telemetry Download

Q462

Mission Operation Task 4: Telemetry Download

In this task you are going to **download a series of telemetry data** from the CubeSat spacecraft. Specifically three types of telemetry: **EPS, temperature and ADCS telemetry**. You must perform this task during the next pass of the CubeSat over the ground station. To accomplish this follow these steps: **EPS Telemetry** Switch to the videoconferencing window. In the MCS terminal and send the following three telecommands (you can copy-paste): **"1: sen_get_eps"** **"1: tm_send_last 2 10"** **"tm_get_last 2"** Based on the output in the MCS terminal answer the question, write your answer in the textbox.

When you are ready click "Next" arrow to advance.

What is the "**vbatt**" (battery voltage) value downloaded from the EPS data?

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Q476

Mission Operation Task 3: Telemetry Download (Continued)

Temperature Telemetry Switch to the videoconferencing window. In the MCS terminal and send the following three telecommands (you can copy-paste): **"1: sen_get_temp"** **"1: tm_send_last 0 10"** **"tm_get_last 0"**

Based on the output in the MCS terminal answer the question, write your answer in the textbox.

When you are ready click "Next" arrow to advance.

What is the "**obc_temp_1**" (temperature of the On-Board Computer) value downloaded from the temperature data?

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Q477

Mission Operation Task 3: Telemetry Download (Continued)

ADCS Telemetry Switch to the videoconferencing window. In the MCS terminal and send the following three telecommands (you can copy-paste): **"1: sen_get_adcs"** **"1: tm_send_last 1 10"** **"tm_get_last 1"**

Based on the output in the MCS terminal answer the question, write your answer in the textbox.

When you are ready click "Next" arrow to advance.

What is the "**mag_x**" (magnetometer) value downloaded from the ADCS data?

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Page Break

Q463 Have you performed the *"***telemetry download***"* operation before? *(Task 3: Question 1 of 7)*

- ○ Yes, several times. (1)
- ○ Yes, a few times. (2)
- ○ No, never (3)

Page Break

Q464 I understood what was the satellite operation being performed. *(Task 3: Question 2 of 7)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q465 Please explain what elements were not clear.

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Q466 Please explain what elements allowed you to understand the operation.

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Page Break

Q467 The **EPS telemetry** output text and data shown when performing the **"telemetry download"** operation resembles the EPS telemetry output text and data of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 3: Question 3 of 7)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q468 Please explain what elements of the output text and data shown do not resemble the operation of a real CubeSat satellite mission.

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Q469 Please explain what elements of the output text and data shown resemble the operation of a real CubeSat satellite mission.

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Q478 The **temperature telemetry** output text and data shown when performing the **"telemetry download"** operation resembles the temperature telemetry output text and data of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 3: Question 4 of 7)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q479 Please explain what elements of the output text and data shown do not resemble the operation of a real CubeSat satellite mission.

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Q480 Please explain what elements of the output text and data shown resemble the operation of a real CubeSat satellite mission.

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Page Break

Q481 The **ADCS telemetry** output text and data shown when performing the **"telemetry download"** operation resembles the ADCS telemetry output text and data of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 3: Question 5 of 7)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q482 Please explain what elements of the output text and data shown do not resemble the operation of a real CubeSat satellite mission.

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Q483 Please explain what elements of the output text and data shown resemble the operation of a real CubeSat satellite mission.

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Q470 The telecommands used in the **"telemetry download"** operation resemble the telecommands used in a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 3: Question 6 of 7)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q471 Please explain what elements of the telecommands do not resemble the operation of a real CubeSat satellite mission.

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Q472 Please explain what elements of the telecommands resemble the operation of a real CubeSat satellite mission.

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Q473 Overall, the **"telemetry download"** operation resembles the operation of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Task 3: Question 7 of 7)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q474 Please explain what overall elements do not resemble the operation of a real CubeSat satellite mission.

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Q475 Please explain what overall elements most resemble the operation of a real CubeSat satellite mission.

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End of Block: Task 3: Telemetry Download

Start of Block: Satellite Operator Task End

Q225

Overall University CubeSat Satellite Honeypot Questions Thank you so much for completing the survey so far. In this last part of the survey, we will ask you to evaluate our CubeSat satellite honeypot as a whole based on all the tasks that you just completed and ask you for some feedback.

End of Block: Satellite Operator Task End

Start of Block: Overall Evaluation Questions

Page Break

Q238 The telemetry output supported by the CubeSat satellite honeypot system resembles the telemetry output operation of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Overall evaluation: Question 1 of 10)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q239 Please explain what elements of the telemetry output shown do not resemble the operation of a real CubeSat satellite mission.

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Q434 Please explain what elements of the telemetry output shown resemble the operation of a real CubeSat satellite mission.

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Q240 The telecommands supported by the CubeSat satellite honeypot system resembles the telecommands of a real CubeSat satellite mission, even if it differs from the missions I have operated in the past. *(Overall evaluation: Question 2 of 10)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q241 Please explain what elements of the telecommands shown do not resemble the operation of a real CubeSat satellite mission.

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Q435 Please explain what elements of the telecommands shown resemble the operation of a real CubeSat satellite mission.

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Q485 I would not be able to distinguish between this CubeSat satellite honeypot system from the real CubeSat satellite mission it is based on. *(Overall evaluation: Question 3 of 10)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q245 Please explain what elements of the satellite honeypot system shown that would allow you to distinguish it from a real satellite mission.

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Q436 Please explain what elements of the satellite honeypot system would not allow you to distinguish it from a different CubeSat real satellite mission.

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Q248 If I did not know this was a honeypot simulation of a CubeSat satellite mission, I would have believed it to be an actual CubeSat satellite mission. *(Overall evaluation: Question 4 of 10)*

- ○ Strongly Disagree (1)
- ○ Disagree (2)
- ○ Neither Agree nor Disagree (3)
- ○ Agree (4)
- ○ Strongly Agree (5)

Q438 Please explain what elements of the satellite honeypot system reveal the fact that it is a simulation.

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Q439 Please explain what elements of the satellite honeypot system makes it a convincing simulation of a CubeSat satellite mission.

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End of Block: Overall Evaluation Questions

Start of Block: Feedback Open-ended Questions

Q249 What features or behaviors would you expect in a real CubeSat satellite operation system that are missing in our simulation? *(Overall evaluation: Question 5 of 10)*

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Q250 What improvements would you suggest to make this CubeSat satellite mission honeypot system more realistic? *(Overall evaluation: Question 6 of 10)*

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Q251 What did you find most convincing about the system, and why? *(Overall evaluation: Question 7 of 10)*

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Q252 If you were tasked with identifying this as a CubeSat satellite mission honeypot, what clues would you rely on? *(Overall evaluation: Question 8 of 10)*

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Q253 Were there any parts of the CubeSat satellite honeypot system that felt unrealistic or inconsistent with your experience? Please describe them. *(Overall evaluation: Question 9 of 10)*

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Q254 Is there anything else you would like us to know? *(Overall evaluation: Question 10 of 10)*

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End of Block: Feedback Open-ended Questions

Start of Block: Does not qualify